

Chapter 8:

Integration Techniques and Improper Integrals

Section 8.4: Trigonometric Substitution

We use trigonometric substitution to tackle integrals that have expressions $\sqrt{a^2 - u^2}$, $\sqrt{a^2 + u^2}$, and $\sqrt{u^2 - a^2}$.

The objective with trigonometric substitution is to eliminate the radical in the integrand.
You do this by using the Pythagorean identities.

$$\cos^2\theta = 1 - \sin^2\theta$$
$$\sec^2\theta = 1 + \tan^2\theta$$
$$\tan^2\theta = \sec^2\theta - 1$$

For example, for $a > 0$, let $u = a\sin\theta$, where $-\pi/2 \le \theta \le \pi/2$. Then

$$\begin{aligned}\sqrt{a^2 - u^2} &= \sqrt{a^2 - a^2\sin^2\theta} \\ &= \sqrt{a^2(1 - \sin^2\theta)} \\ &= \sqrt{a^2\cos^2\theta} \\ &= a\cos\theta.\end{aligned}$$

Note that $\cos\theta \ge 0$, because $-\pi/2 \le \theta \le \pi/2$.

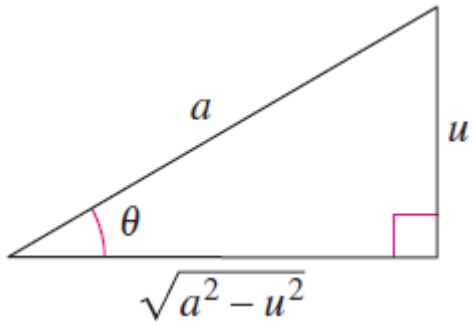
Trigonometric Substitution ($a > 0$)

1. For integrals involving $\sqrt{a^2 - u^2}$, let

$$u = a \sin \theta.$$

Then $\sqrt{a^2 - u^2} = a \cos \theta$, where

$$-\pi/2 \le \theta \le \pi/2.$$

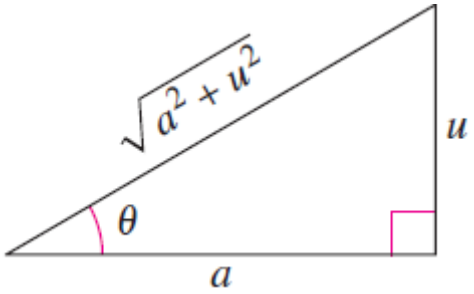


2. For integrals involving $\sqrt{a^2 + u^2}$, let

$$u = a \tan \theta.$$

Then $\sqrt{a^2 + u^2} = a \sec \theta$, where

$$-\pi/2 < \theta < \pi/2.$$

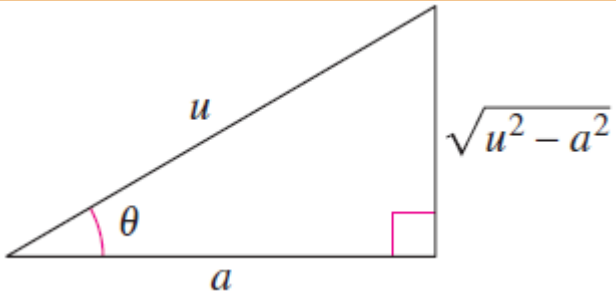


3. For integrals involving $\sqrt{u^2 - a^2}$, let

$$u = a \sec \theta.$$

Then

$$\sqrt{u^2 - a^2} = \begin{cases} a \tan \theta \text{ for } u > a, \text{ where } 0 \le \theta < \pi/2 \\ -a \tan \theta \text{ for } u < -a, \text{ where } \pi/2 < \theta \le \pi. \end{cases}$$



EXAMPLE

Find $\int \sqrt{4 - x^2}\ dx.$

SOLUTION

$$I = 2\arcsin\left(\frac{x}{2}\right) + \frac{x}{2}\sqrt{4 - x^2} + C$$

EXAMPLE

Find $\int \frac{dx}{(4x^2+1)^{\frac{3}{2}}}$.

$$I = \frac{x}{\sqrt{4x^2 + 1}} + C$$

EXAMPLE

Find $\int_{\sqrt{3}}^{2} \frac{\sqrt{x^2-3}}{x^2}\ dx.$

$$I = \frac{1}{2}\ln(3) - \frac{1}{2}$$

EXAMPLE

Completing Square

Find $\int \frac{x^2}{\sqrt{5-4x-x^2}}\,dx.$

$$I = \frac{17}{2}\arcsin\left(\frac{x+2}{3}\right) - \frac{x+2}{2}\sqrt{5-4x-x^2} + 4\sqrt{5-4x-x^2} + C$$

EXAMPLE

Find $\int \sqrt{9-x^2}\ dx.$

Find $\int x\sqrt{9-x^2}\ dx.$

Find $\int x^2\sqrt{9-x^2}\ dx.$

Find $\int x^3\sqrt{9-x^2}\ dx.$

EXAMPLE **Think about them**

$$\int \frac{x^2 + 1}{(x^2 - 2x + 2)^2}\, dx$$

$$\int x\sqrt{1 - x^4}\, dx$$

$$\int_0^{\pi/2} \frac{\cos t}{\sqrt{1 + \sin^2 t}}\, dt$$

Solve exercises of section 8.4.

Start with the suggested problems mentioned in the syllabus.